Compounds of Transitional elements

121. The compound insoluble in water is
(a) Mercurous nitrate
(b) Mercuric nitrate
(c) Mercurous chloride
(d) Mercurous perchlorate

122. Which is an amphoteric oxide
(a) ZnO (b) CaO
(c) BaO (d) SrO

123. What is the magnetic moment of $[FeF_6]^{3-}$
(a) 5.92 (b) 5.49
(c) 2.32 (d) 4

124. How H_2S is liberated in laboratory
(a) $FeSO_4 + H_2SO_4$
(b) $FeS + dil.\ H_2SO_4$
(c) $FeS + \text{conc.}\ H_2SO_4$
(d) Elementary H_2+ elementary S

125. The spin magnetic moment of cobalt in the compound $Hg[Co(SCN)_4]$ is
(a) $\sqrt{3}$ (b) $\sqrt{8}$
(c) $\sqrt{15}$ (d) $\sqrt{24}$

126. In which of these processes platinum is used as a catalyst
(a) Oxidation of ammonia to form HNO_3
(b) Hardening of oils
(c) Production of synthetic rubber
(d) Synthesis of methanol

127. Iron is dropped in dil. HNO_3, it gives
(a) Ferric nitrate
(b) Ferric nitrate and NO_2
(c) Ferrous nitrate and ammonium nitrate
(d) Ferrous nitrate and nitric oxide

128. CrO_3 dissolves in aqueous $NaOH$ to give
(a) CrO_4^{2-} (b) $Cr(OH)_3^-$
(c) $Cr_2O_7^{2-}$ (d) $Cr(OH)_2$

129. KI and $CuSO_4$ solution when mixed, give
(a) $CuI_2 + K_2SO_4$
(b) $Cu_2I_2 + K_2SO_4$
(c) $K_2SO_4 + Cu_2I_2 + I_2$
(d) $K_2SO_4 + CuI_2 + I_2$

130. When Cu reacts with $AgNO_3$ solution, the reaction takes place is
(a) Oxidation of Cu
(b) Reduction of Cu
(c) Oxidation of Ag
(d) Reduction of NO_3^-

131. By annealing, steel
(a) Becomes soft
(b) Becomes liquid
(c) Becomes hard and brittle
(d) Is covered with a thin film of Fe_3O_4

132. Which of the following is more soluble in ammonia
(a) $AgCl$
(b) $AgBr$
(c) AgI
(d) None of these

133. Potassium permagnate works as oxidising agent both in acidic and

basic medium. In both state product obtained by $KMnO_4$ is respestively

(a) MnO_2^- and Mn^{3+}
(b) Mn^{3+} and Mn^{2+}
(c) Mn^{2+} and Mn^{3+}
(d) MnO_2 and Mn^{2+}
(e) Mn^{2+} and MnO_2

134. Which of the followign is the green coloured powder produced when ammonium dichromate is used in fire works

(a) Cr (b) CrO_3
(c) Cr_2O_3 (d) $CrO(O_2)$

135. Which compound does not dissolve in hot dilute HNO_3

(a) HgS (b) CuS
(c) PbS (d) CdS

136. The least stable oxide at room temperature is

(a) ZnO (b) CuO
(c) Sb_2O_3 (d) Ag_2O

137. Which of the following pari of elements cannot form an alloy

(a) $Zn - Cu$ (b) $Fe - Hg$
(c) Fe, C (d) Na, Hg

138. Which of the following shows dimerisation

(a) $HgCl_2$ (b) B_2H_6
(c) $TiCl_4$ (d) SO_2

139. Which of the following is also known as "Fools gold"

(a) Wurtzite
(b) Iron pyrites
(c) Chalcosite
(d) Silver glance

140. Which one of the following is highest melting halide

(a) AgCl (b) AgBr
(c) AgF (d) AgI